

9708/31/m/j/17

1-B:

A – Higher prices for hotel rooms at holiday periods is representative of “price discrimination”

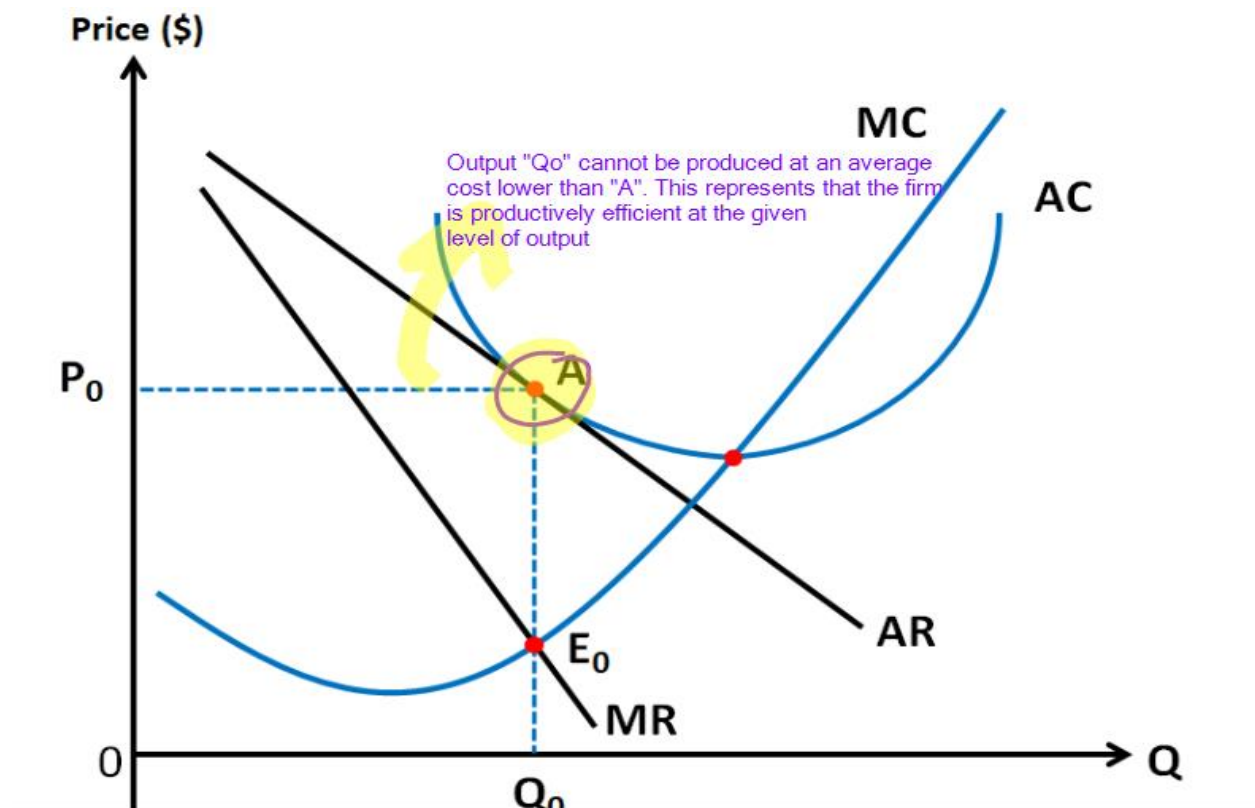
B – Noise pollution experienced near major airports represents external costs and since they are not internalized by the competitive free market, they give existence to “market failure”. This is why, this becomes correct.

C and D: Inflation and poverty are macro-economic issues.

2-D:

A , B, C: The removal of all externalities , elimination of all divergences between private and social costs, and the equality between price and marginal cost in all industries represent “allocative efficiency”

D: In the short-run, for productive efficiency to be achieved, a given output must be produced at the minimum average cost possible. This can be seen in the diagram below.



3A

The table shows the expected costs and benefits from four government projects. The government can afford only one project.

Which project should the government choose?

SB	private benefits (\$m)	external benefits (\$m)	private costs (\$m)	external costs (\$m)	SC	Net SB
240	40	200	60	70	130	110
220	60	160	100	20	120	100
340	100	210	100	120	220	120
240	150	90	120	140	260	-20

Net social benefit = $SB - SC$

Highest net social benefit (option A)

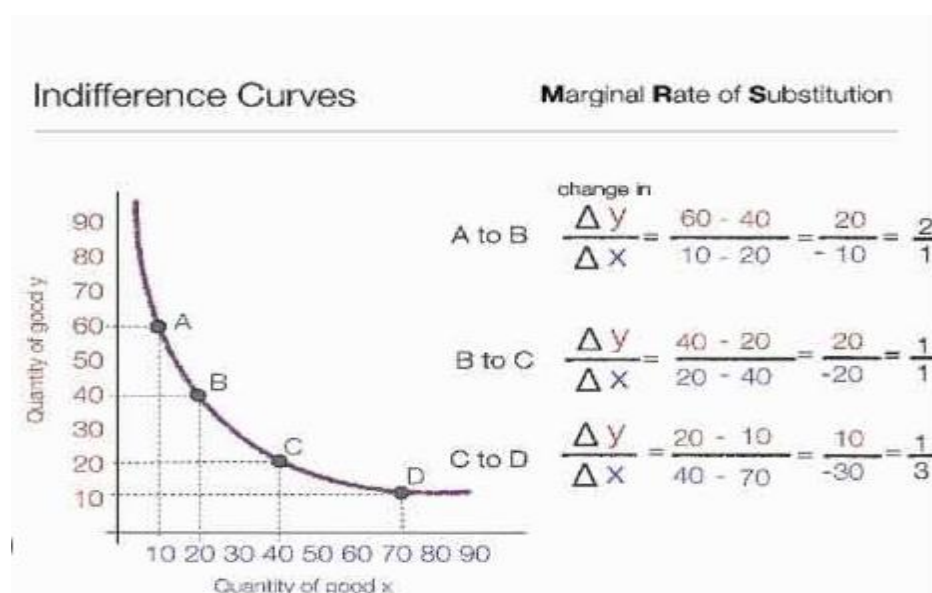
$$* SB = PB + EB$$

$$* SC = PC + EC$$

4-B:

A – The indifference curves for inferior goods do not show a higher satisfaction on lower curves. **This is because, the further the indifference curve is from the origin, the greater will be the satisfaction that it will yield.**

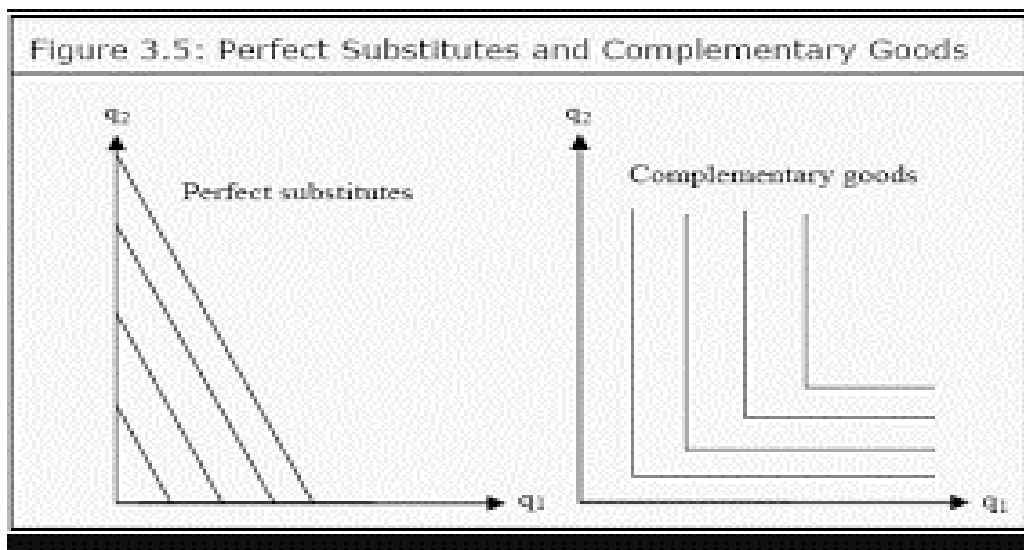
B – MRS represents the rate at which a consumer is willing to give up on a good in return for another. The indifference curve for normal goods show a diminishing marginal rate of substitution.



- ➔ Initially, the consumer operating at point A, had a higher quantity of good Y, as a result of which diminishing marginal utility for good Y was faced. This means that he would be willing to give up more of good Y for some quantity of good X.
- ➔ Furthermore, now when the consumer operates at point C, a higher quantity of good X is already being purchased, which represents that diminishing marginal utility of good X is being faced. As a result of which, the consumer would be willing to give up less of good Y for some of X.
- ➔ This is because of the law of diminishing marginal utility, that the convex shaped indifference curve shows **diminishing marginal rates of substitution. This is why, this becomes correct.**

C and D:

The shapes of indifference curves for perfect substitutes and complements are shown below:



5-B:

- Since the utility that the consumer derives from the last dollar (\$1) he spends on the good is 4 utils, a rational consumer would obtain utility of 40 utils, when \$10 is spent (10×4)



The utility he derives from the last \$ he spends on every good is 4 units.

Assuming the marginal utility of money is constant, which quantity will he purchase if the price of the good is \$10?

- A 4 kilos
 - **B 5 kilos**
 - C 6 kilos
 - D 7 kilos
- 6-B:

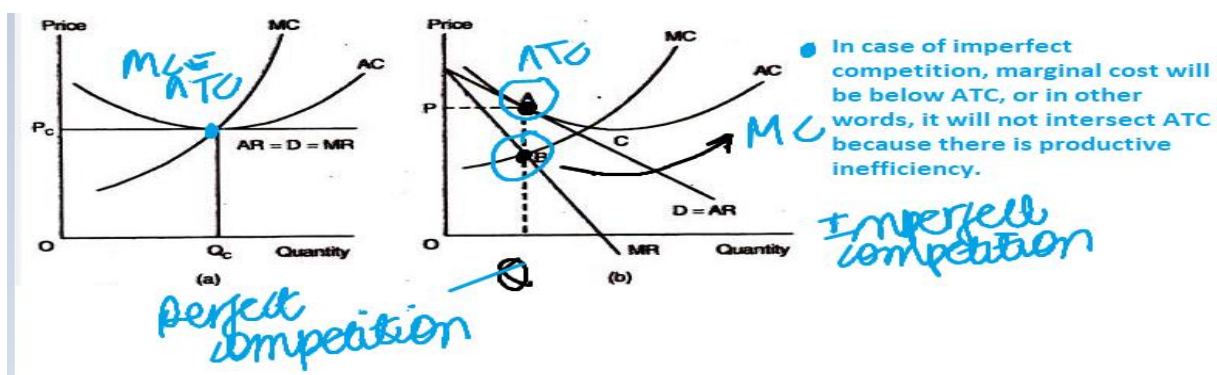
A) Horizontal integration occurs when a company acquires or merges with another company in the same industry at the same production stage. In this case, a bank taking over a life assurance company is not an example of horizontal integration, as they are in different industries (banking and insurance).

B) Lateral integration occurs when a company acquires or merges with another company in a related industry or market. A bank taking over a life assurance company is an example of lateral integration, as both companies operate in the financial services sector and can complement each other's services.

C) Vertical backwards integration occurs when a company acquires or merges with a supplier or a company in an earlier stage of production.

D) Vertical forwards integration occurs when a company acquires or merges with a distributor or a company in a later stage of production.

7-C: The following diagram explains why key c is correct



A – In case of imperfect competition, supply does not always have to be elastic in the long-run; rather there is no significance of supply curve in imperfect competition in the long-run.

B – In case of imperfect competition, demand does not always have to be inelastic. This is because, monopolistically competitive firms face “elastic demand”

D – Average revenue will not be below marginal revenue; rather, it will be above Marginal revenue.

8-C:

- **The correct answer is option C. Output level “C” represents that we are operating at an allocatively efficient equilibrium where “MC=AR”**
- **Output level D represents that we are operating at a normal profit equilibrium where ATC=AR.**
- **Output level B represents that we are operating at a revenue maximizing equilibrium where MR=0**
- **Output level A represents that we are operating at a profit maximizing equilibrium where MC=MR**

9-D:

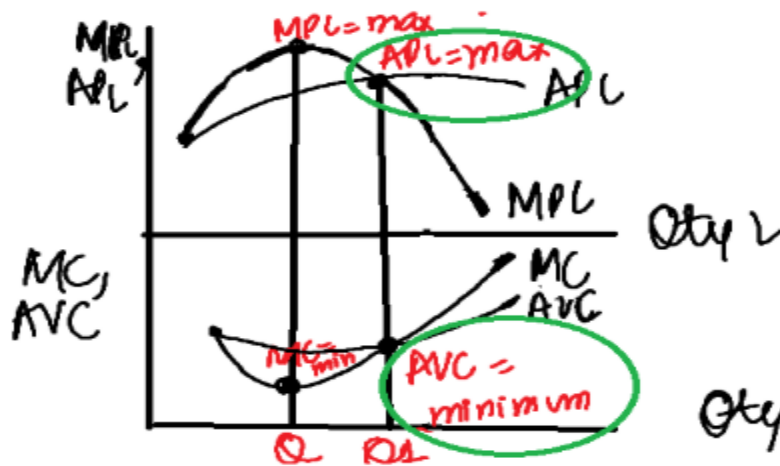
A) This option is incorrect because the **expiration of copyrights** would actually make it easier for a small firm to compete, as they could use the product design without infringing on the large firm's intellectual property rights.

B) This option is **incorrect** because if customers want individual product designs, it would actually be easier for a small firm to compete, as they can more easily adapt to customization requests.

C) This option is incorrect because **a low entry cost** would actually make it easier for a small firm to compete, as they wouldn't need to invest a lot of resources to enter the market.

D) When the **industry's product is standardized**, it makes it harder for a small firm to compete against large firms, as large firms can produce at a lower cost due to economies of scale, invest more in marketing, and make their product more visible to customers. This is why, this becomes correct.

10-A:



A – When average product of labor is at its maximum, that is when the “labor productivity” is at its highest, and since labor contributes the least to the variable cost and the most to the output, “average variable cost” is at its lowest. This is why, key A, becomes correct. See the diagram above

B- Marginal cost is at its minimum when Marginal product of labor is at its maximum.

C – Again, similar to option B, marginal product is at its maximum when marginal cost is at its minimum.

D – We can not identify total profit based on this information

11-B:

- ➔ Since the dominant firm in an oligopoly is setting up a research and development institute to undertake product development, this represents that the firm is engaging in **product differentiation**(which is a form of **non-price competition**) to **differentiate itself from other rivals**. This gives us a hint that we are operating in a non-collusive oligopoly (therefore any option that has “collusion” in it becomes incorrect).
- ➔ Moreover, because of this **barriers to entry** will be created too, as R&D can create complex technologies that are difficult for new entrants to understand and replicate. Additionally, R&D requires significant investments in personnel, equipment, and resources, making it **challenging for new entrants to bear these costs**.

Note: Price leadership is a market situation where a dominant firm sets the price for a product or service, and other firms in the industry follow its lead. The dominant firm, often the market leader or a firm with significant market power, sets the price, and other firms adjust their prices accordingly. There is no occurrence of price leadership here.

This represents why key B will be correct.

12- D:

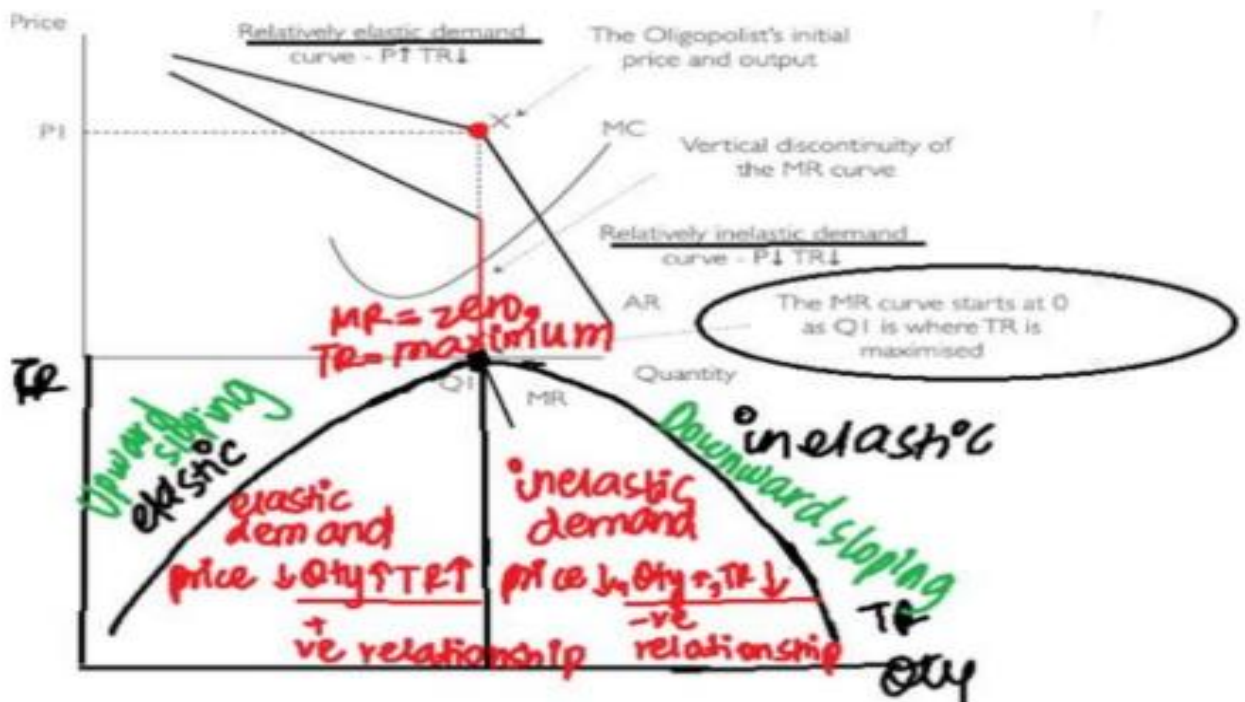
A – The aim of satisficing occurs when it is aimed to earn minimum, acceptable level of profits that are enough to keep shareholders happy.

B – The establishing of a legal requirement represents a legal barrier.

C – An essential feature of a contestable market is a potential threat of competition arising from weak barriers to entry and exit.

D - The idea of persuasion is an essential feature of Nudge Theory because Nudge Theory aims to influence people's behavior in a **subtle, indirect way, rather than through direct coercion or force**. The goal of Nudge Theory is to "nudge" people toward making better choices, and persuasion is a key mechanism for achieving this goal. This is why, this becomes correct.

13-D:



This explains why, key D, will be correct.

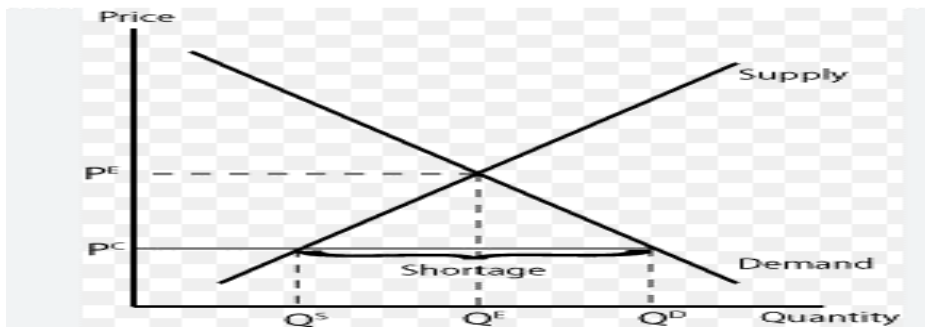
14-D:

A – When because of a maximum price legislation, prices are lowered, the level of inflation is kept in control.

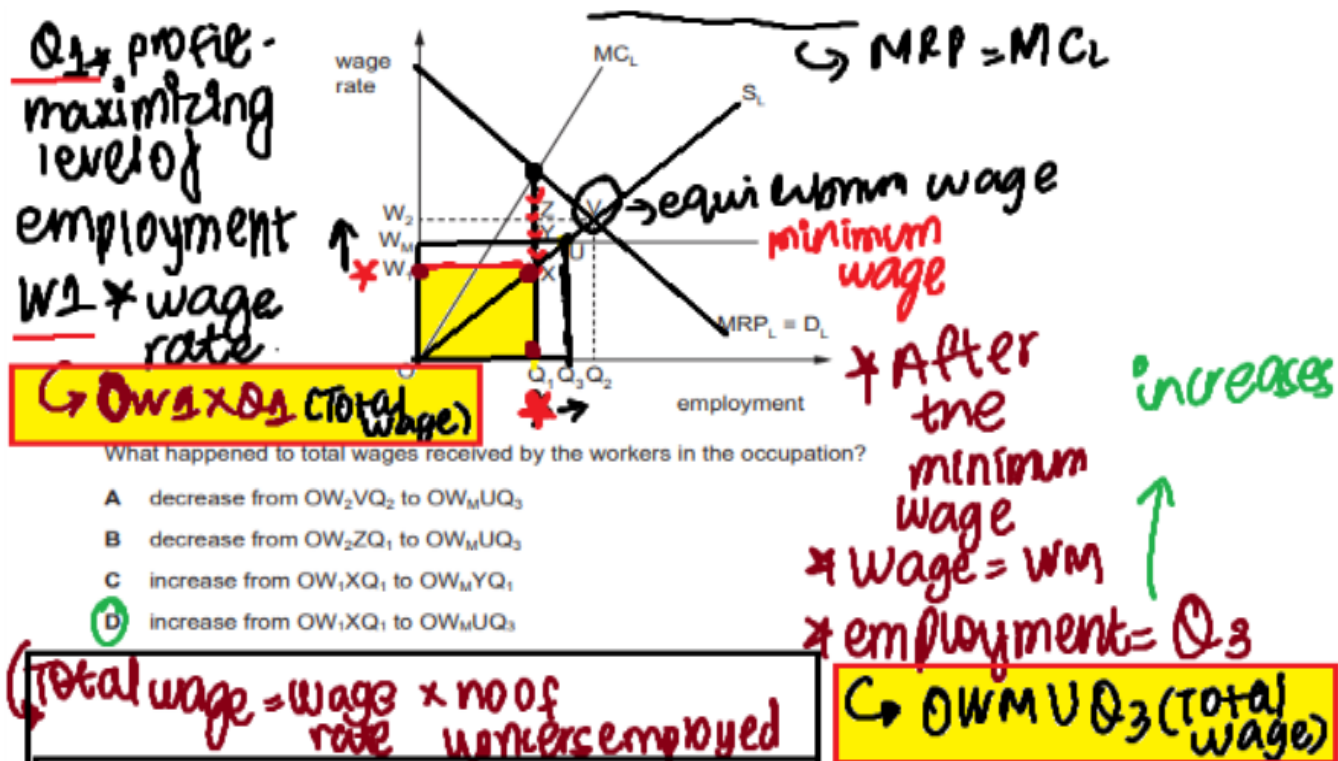
B – Maximum price regulations, imposed at the competitive price line where $MC=AR$, prevent monopolies from earning high levels of profits at the expense of exploitation of consumers.

C – This is incorrect because a maximum price legislation does not result in surpluses.

D – When prices are lowered through maximum prices, quantity demanded extends relative to quantity supplied which contracts, which leads to shortages. This can be seen in the diagram below:-



15-D:



16-A:

A – This is correct because “**economic rent**” represents the extra payment made to a factor of production over and above the minimum level they need in order to work, or in other words, transfer earnings.

B – Normal profits represent “**zero-economic profits**” that are made when Average revenue is equal to Average total cost.

C – **Opportunity cost** represents the real cost of choosing one option and not the another i.e. it is the value of the next best alternative foregone.

D – This is incorrect because **transfer earnings** represent the minimum income a worker needs in order to supply their labor.

17-A:

->The higher the value of gini- coefficient is, the less equal will be the distribution of income.

-> Similarly, the lower the value of gini-coefficient, the more equal will be the distribution of income.

A – It can be said that income is distributed more equally in Denmark because it has a lower gini-coefficient of 24.7, as compared to France which has that of 32.7. This is why, this option becomes correct.

B – Income is distributed less equally in Namibia than in Zambia because Namibia has a relatively higher gini-coefficient.

C – Based on the information of gini-coefficient, we cannot deduce any information about income per head.

D – Based on the information of gini-coefficient, we cannot deduce how many workers live below the poverty line.

18-A:

The carrying capacity of ships has fallen from **17.3 million tonnes to 15.2 million tonnes by 2.1 million tonnes.**

Similarly, the number of ships fell by a relatively higher number of 212 from 712 ships to 500 ships.

Previously, the average size of each ship was: $17.3/712$ i.e. 0.0243 million tonnes (calculated by total carrying capacity/ no of ships)

Now after the number of ships fell by a higher percentage, and when because of it, the total carrying capacity fell by a relatively lesser percentage, this implies that the average size of each ship rose.

This can be numerically confirmed too, as now the average size of each ship is 0.304 million tonnes ($15.2/500$)

This is why, option A, becomes correct

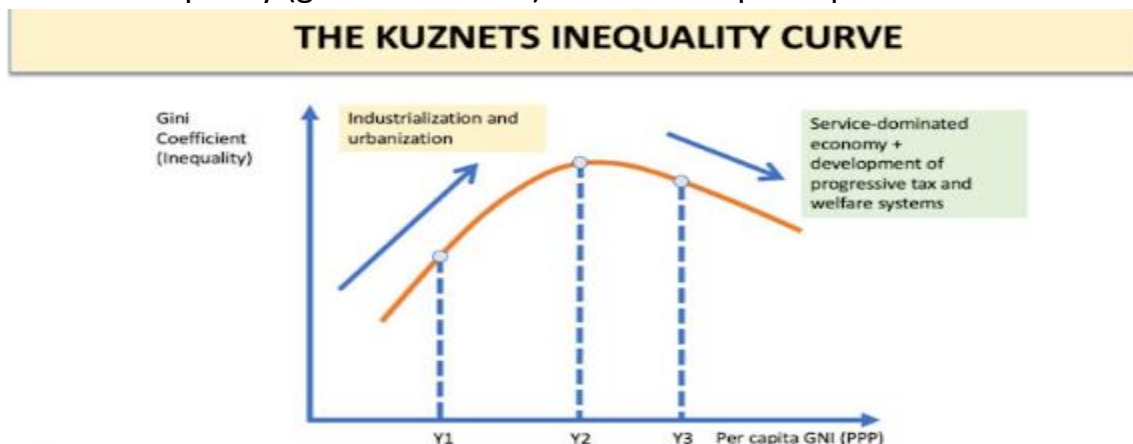
B – The number of voyages made by British registered ships did not fall

C – Moreover, we cannot deduce anything related to productivity based on the information on carrying capacity and number of ships.

D – The capacity of the British registered ships did not fall continuously; instead, it rose in between 2011-2012.

19-A:

A – This is correct because only Kuznets curve shows the relationship between income inequality (gini co-efficient) and income per capita



B – laffer curve shows the relationship between tax rate and tax revenue.

C – Lorenz Curve is a graphical representation of income inequality or wealth inequality. It plots percentiles of the population on the horizontal axis according to income or wealth and plots cumulative income or wealth on the vertical axis .

D – Philips curve shows the negative relationship between inflation and unemployment.

20-D:

A – A fall in savings will bring about a rise in consumption, and subsequently Aggregate demand will increase, as a result of which, there will be a rise in the level of national income in an economy.

B – A fall in the marginal propensity to import represents that more of the additional income earned is spent on domestic goods rather than imported goods. And this rise in consumption would subsequently bring about an increase in Aggregate demand, as a result of which, there will be a rise in the level of national income in an economy.

C – Again, a rise in average propensity to consume indicates that a greater proportion of the overall income earned by households is spent. And this rise in consumption would subsequently bring about an increase in Aggregate demand, as a result of which, there will be a rise in the level of national income in an economy

D- A reduction in government expenditure (G) leads to a fall in Real National Income (RNI). When G decreases, aggregate demand falls, leading to a reduction in income for individuals and businesses. This reduction in income triggers a downward multiplier effect, as households and businesses reduce their spending, leading to a further decrease in aggregate demand and subsequently, a fall in RNI.

Numerical Example:

Let's say the government reduces its expenditure by 100 ($\Delta G = -100$). The multiplier is -5.

1. Initial reduction in government expenditure: G falls by 100

2. Multiplier effect: The reduction in G leads to a reduction in aggregate demand and is multiplied by the multiplier ($k = -5$).

3- Fall in Real National Income (RNI): $\Delta RNI = -500$

This is why, key D, becomes correct.

21-C:

- ➔ When the government borrows from the central bank, it injects money into the economy, increasing the money supply. This is because the central bank creates new money to lend to the government, which is then spent into the economy.

The money supply increases by more than the initial increase in government expenditure because of the money multiplier effect. Here's how it works:

1. The government borrows 100 from the central bank.
2. The central bank creates new money to lend to the government, increasing the money supply by 100.
3. The government spends the 100, which becomes someone else's income.
4. That someone else deposits the 100 into their bank.
5. The bank can now lend a portion of that deposit (let's say 80) to someone else.
6. That someone else spends the 80, which becomes someone else's income.
7. This process continues, with each bank lending a portion of the previous deposit, creating new money and increasing the money supply.

The money multiplier effect is the ratio of the overall increase in money supply to the initial increase in government expenditure. In this example, if the money multiplier is 5, the money supply would increase by 500 (5×100)

22-A:

For inflation to be most likely present, we need to see for a year which has the highest positive output gap.

- ➔ In this case, we had a highest positive output gap of 5% of potential gdp in 1996, which makes key A the answer.
- ➔ Similarly, in 1970, there was no output gap(0% of potential gdp)
- ➔ Moreover in years 1982 and 1992, there was negative output gap, which means that there must have been deflation.

23-A:

Country A has the highest degree of development. This is because, the percentage of population in rural areas is the least.

Moreover, the average years of schooling and life expectancy at birth are also the highest, which are important qualitative aspects of standard of living.

Similarly, the gdp per capita is also the highest.

24-B:

A recession is a period of economic decline, typically defined as a decline in gross domestic product (GDP) for two or more consecutive quarters.

- ➔ **The first recession occurred in between 2010-2011, when there was a fall in real gdp per capita for three quarters.**
- ➔ **The second occurred in 2012, when there was a fall in real gdp per capita for three quarters**
- ➔ **The third occurred in 2014, when there was a fall in real gdp per capita for two quarters.**

25-B:

The **labor force** is the **total number of people who are either employed or unemployed, but actively seeking work.** The formula is: **Labor Force = Employed + Unemployed**

Where:-**Employed** refers to people who have a job, whether full-time or part-time. -**Unemployed** refers to people who are actively seeking work but do not have a job.

- The **working age population**, refers to the total number of people in a country who are eligible to work. This population is divided into two main categories:

1. Labor Force

2. Not in Labor Force

The equation is:

Working Age Population = Labor Force + Not in Labor Force

- Labor Force includes people who are:

- **Employed (working)**
- **Unemployed (actively seeking work)**

- Not in Labor Force includes people who are:

- Not working and not seeking work (e.g., students, retired, stay-at-home parents, disabled)

This explains why option B is correct

26-C:

This mcq is not available

27-A:

A) - Lowering exchange rate → cheaper exports → demand extends by a relatively higher percentage when it is elastic → export revenues rise → net injections into the circular flow → increased Aggregate demand → higher production → a rise in demand for labor → more employment.

- Lowering domestic interest rates → cheaper borrowing → increased investment → higher Aggregate demand → higher production → a rise in demand for labor → more employment

- Combination → multiplier effect → even more employment → reduced unemployment

Since this will help to battle cyclical unemployment arising from a deficiency of Aggregate demand, option A, becomes correct.

B) - Lowering exchange rate → mentioned above

- Increasing direct taxation → reduced disposable income → lower consumption → lower Aggregate demand → less employment

- Combination → conflicting effects → no clear reduction in unemployment

C) Raising exchange rate → more expensive exports → the demand for exported goods contract by a relatively higher percentage when it is elastic → export revenues fall → fall in AD → lower production → fall in demand for labor → less employment

- Raising domestic interest rates → more expensive borrowing → reduced investment → lower Aggregate demand → less employment

- Combination → amplified negative effects → increased unemployment

D) Raising exchange rate → mentioned above

- Increasing direct taxation → mentioned above

- Combination → cumulative negative effects → increased unemployment

28-A:

-> By keeping tax rates and benefit rates unchanged, the government is allowing automatic stabilizers to work. **This helps to stabilize output and employment without the need for discretionary policy changes.**

Automatic stabilizers are features of the tax and transfer system that automatically increase government spending or reduce taxes when output falls, and reduce government spending or increase taxes when output rises.

In a recession, tax revenues automatically fall, and spending on benefits like unemployment insurance automatically increases, helping to cushion the downturn. In a boom, tax revenues automatically rise, and spending on benefits falls, helping to slow down the economy. **By not changing tax and benefit rates, the government is allowing these automatic stabilizers to work, helping to stabilize output and employment over the business cycle.**

29-B:

A) A decrease in government spending would reduce aggregate demand and worsen the economic downturn.

B) A decrease in unemployment represents that individuals will have more incomes in hand, which will stimulate consumption, thereby boosting Aggregate demand, subsequently helping the economy to recover.

C) An increase in income tax would reduce disposable income, leading to lower consumption and subsequently AD, thereby worsening the economic downturn.

D) An increase in interest rates would make borrowing more expensive, reducing investment, and worsening the economic downturn.

30-C

Cutting personal income taxes increases disposable income, leading to higher consumption and aggregate demand. This is likely to stimulate economic growth and reduce unemployment, as businesses respond to increased demand by hiring more workers.

The other options are not the main macroeconomic objectives:

A) Cutting income taxes is likely to worsen the balance of payments position, as higher disposable incomes lead to increased imports.

B) Cutting income taxes will likely increase the budget deficit, as the government reduces its revenue.

D) Cutting income taxes is likely to increase demand-pull inflation, as when Aggregate demand outpaces Aggregate Supply, there will be shortages which will subsequently lead to higher prices.